Local Setup Guide

AI Business Card CRM — MSTinternship Project

Step 1 — Prerequisites

Install all of the following before proceeding:

Tool	Purpose	Where to Get It
Python 3.11+	Runs Django	python.org
PostgreSQL	Database	postgresql.org
Redis	Background task queue	redis.io
Ollama	Runs AI models locally	ollama.com
Git	Clone the repository	git-scm.com

Step 2 — Clone the Repository

```
git clone https://github.com/Rehansh26/MSTinternship-Rehansh-Shrivastava.git
```

```
cd MSTinternship-Rehansh-Shrivastava
```

Step 3 — Create a Virtual Environment

Mac / Linux:

```
python3 -m venv venv
source venv/bin/activate
```

Windows:

```
python -m venv venv
venv\Scripts\activate
```

Step 4 — Install Dependencies

```
pip install -r requirements.txt
```

Step 5 — Set Up the Database (PostgreSQL)

Open pgAdmin or psql and run:

```
CREATE DATABASE crm_db;
```

Step 6 — Configure Environment Variables

A **.env.example** file is included in the repository with all the required variable names. Copy it and fill in your own values:

Mac / Linux:

```
cp .env.example .env
```

Windows:

```
copy .env.example .env
```

Then open **.env** in any text editor and fill in your values:

```
SECRET_KEY=your-secret-key-here
DB_NAME=crm_db
DB_USER=postgres
DB_PASSWORD=your-postgres-password
DB_HOST=localhost
DB_PORT=5432
RAZORPAY_KEY_ID=rzp_test_xxxxxxxxxxxx
RAZORPAY_KEY_SECRET=your-razorpay-secret-here
```

Note: Razorpay keys are optional for local testing — the app works without them using a mock fallback.

Step 7 — Pull the AI Models (one-time, large downloads)

```
ollama pull llama3.2-vision   # reads business card images (~8 GB)
ollama pull llama3.2          # chat assistant (~2 GB)
ollama pull nomic-embed-text  # semantic search (~300 MB)
```

Step 8 — Run Database Migrations

```
python manage.py migrate
python manage.py createsuperuser
```

The createsuperuser command sets your login username and password for the app.

Step 9 — Start All 4 Services

Open 4 separate terminal tabs, all inside the project folder with the venv activated:

Tab 1 — Django web server:

```
python manage.py runserver
```

Tab 2 — Celery worker (processes card scans in background):

```
celery -A card_manager worker --loglevel=info
```

Tab 3 — Redis (if not running as a system service):

```
redis-server
```

Tab 4 — Ollama (if not running as a system service):

```
ollama serve
```

Step 10 — Open the App

Go to **http://127.0.0.1:8000** in your browser and log in with the superuser credentials you created in Step 8.

Quick Checklist

■ PostgreSQL running and crm_db database exists

■ .env file created from .env.example with correct values

■ pip install -r requirements.txt completed with no errors

■ All 3 Ollama models downloaded

■ python manage.py migrate ran successfully

■ All 4 services running (Django, Celery, Redis, Ollama)

Common Errors

Error	Fix
connection refused on DB	PostgreSQL is not running, or wrong password in .env
connection refused on Redis	Start Redis with: redis-server
Card stuck on "AI is scanning"	Celery is not running, or Ollama is not running
ModuleNotFoundError	Venv not activated, or pip install not done